
HOMEWORK 2 (MODERN OPTIMIZATION ALGORITHMS, MATH 485)

Due date: See class webpage on Canvas.

Unless otherwise stated, $\mathcal{H}$ is a real Hilbert space with inner product $\langle \cdot | \cdot \rangle$ and induced norm $\| \cdot \|$.

DEFINITIONS A set $C \subset \mathcal{H}$ is **bounded** if there exists $M \in [0, +\infty[$ such that, for all $c \in C$,

$$\|c\| \leq M.$$

The **Minkowski sum** of two sets $A, B \subset \mathcal{H}$ is

$$A + B = \{a + b \mid a \in A \text{ and } b \in B\}.$$

FACT Let $T: \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be a bounded linear operator between real Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$. Using the definition of $\|T\|$ from class, it can be shown that ($\forall x \in \mathcal{H}_1$),

$$\|Tx\|_{\mathcal{H}_2} \leq \|T\| \|x\|_{\mathcal{H}_1} \tag{1}$$

Proof. If $x = 0 \in \mathcal{H}_1$, then since every linear operator maps 0 to 0, the claim (1) is satisfied since $\|Tx\| = 0 \leq 0 = \|T\| \cdot \|x\|$. Now suppose $x \neq 0$. Then $\|x\| > 0$. So, since we can pull scalars out of linear operators, we can then use positive homogeneity of the norm to find

$$\|Tx\|_{\mathcal{H}_2} = \left\| T \left(\frac{\|x\|}{\|x\|} x \right) \right\| = \left\| \|x\| T \left(\frac{1}{\|x\|} x \right) \right\| = \|x\| \left\| T \left(\frac{1}{\|x\|} x \right) \right\| \leq \|x\| \sup_{\substack{y \in \mathcal{H} \\ \|y\| \leq 1}} \|Ty\| = \|T\| \|x\|$$

where the final inequality is due to the fact that “ $y = x/\|x\|$ ” satisfies $\|y\| = \|x/\|x\|\| = (1/\|x\|)\|x\| = 1 \leq 1$, so the supremum in $\|T\|$ upper bounds $\|T(x/\|x\|)\|$. $\square$

QUESTIONS

1. Let $0_{\mathcal{H}}$ be the zero element of $\mathcal{H}$ (i.e., the additive identity satisfying ($\forall x \in \mathcal{H}$) $x + 0_{\mathcal{H}} = x$ and $0_{\mathbb{R}} \cdot x = 0_{\mathcal{H}}$). Show that, for every $x \in \mathcal{H}$, $\langle x | 0_{\mathcal{H}} \rangle = 0_{\mathbb{R}}$.
2. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be real Hilbert spaces, let $C \subset \mathcal{H}_1$ be bounded, and let $T: \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be a bounded linear operator. Show that $T(C)$ is bounded. (hint: use the Fact above).
3. If C_1 and C_2 are bounded sets, show that $C_1 + C_2$ is bounded.

4. Let $x, y \in \mathbb{R}^n$ be such that every component of x is 1, and the components of y are given by

$$y_i = \begin{cases} 1 & \text{if } i \text{ is odd} \\ -1 & \text{if } i \text{ is even.} \end{cases} \quad (2)$$

For instance, if $n = 4$, $y = [1, -1, 1, -1]$ and for $n = 5$, $y = [1, -1, 1, -1, 1]$ and so on. Determine how the angle between x and y changes with dimension n : Is their angle acute, obtuse, or orthogonal?

EC: Conduct the same exercise for determining the angle between y (same as above) and $z \in \mathbb{R}^n$, where $z_1 = 1$ and for every $i \geq 2$, $z_i = (\frac{1}{i} + \frac{1}{i-1})$. Is their angle acute, obtuse, or orthogonal for finite n values? What happens to their angle as $n \rightarrow \infty$?

5. Prove that $\|\cdot\| = \sqrt{\langle \cdot | \cdot \rangle}$ is a norm.